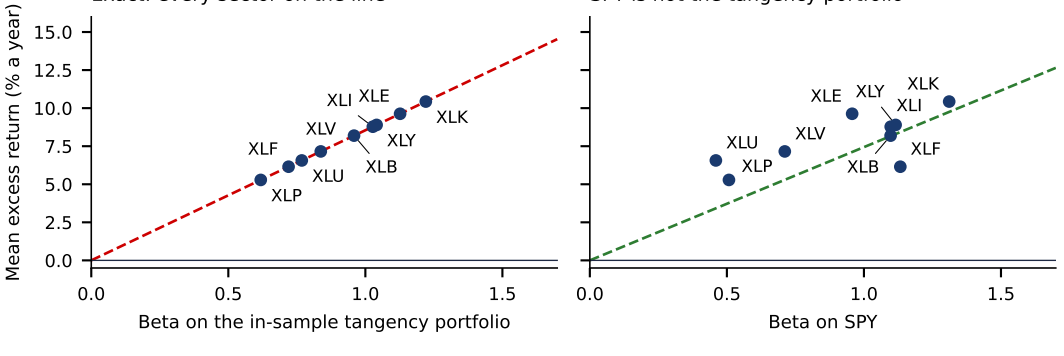
Exact: every sector on the line
SPY is not the tangency portfolio
Mean excess return (% a year)
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.0
0.5
1.0
1.5
XLI
XLE
XLK
XLV
XLY
XLF
XLB
XLU
XLP
Beta on the in-sample tangency portfolio
Beta on SPY
SML of the tangency portfolio
Sector ETFs
SML of SPY